

Examination on Java Programming

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Consider the following classes implementing ephemeral stacks:

```
public class Stack<Key extends Comparable<Key>> {
    protected Cell<Key> top;
    public Stack () { top = null; }
    public void push (final Key k) {
        top = new Cell<Key>(k,top); }
}
```

```
public class Cell<Key extends Comparable<Key>> {
    protected Key value;
    protected Cell<Key> next;
    public Cell (final Key key, final Cell<Key> cell) {
        value = key; next = cell; }
}
```

Extend the class `Stack<Key>` with two methods defined as follows.

1. The method `find_front` whose signature is

```
public int find_front (final Key k) throws NotFound;
```

performs a linear search in **this**: if the key `k` is absent, then the exception `NotFound` (to be defined) is thrown; otherwise, the index of the first cell containing `k` is returned (by definition, the top of the stack has index 0) and is moved to the top. This heuristics improving the efficiency of linear search when it is used often is called *move to front*.

2. The method `find_swap` whose signature is

```
public int find_swap (final Key k) throws NotFound;
```

performs the same operations as `find_front`, except that the first cell containing `k` (if any) is swapped with the previous one (if any).

The main class for testing is

```
public class EphObj {
    public static void main (String[] args) {
        Stack<Integer> s = new Stack<Integer>();
        s.push(5); s.push(3); s.push(1); s.push(0);
        s.print(); // 0 1 3 5
        try { System.out.println(s.find_front(5)); } // 3
        catch (NotFound x) {}
        s.print(); // 5 0 1 3
        try { System.out.println(s.find_front(5)); } // 0
        catch (NotFound x) {}
        s.print(); // 5 0 1 3
        try { System.out.println(s.find_front(1)); } // 2
        catch (NotFound x) {}
        s.print(); // 1 5 0 3
        try { System.out.println(s.find_front(7)); }
        catch (NotFound x) { System.out.println("Not found."); }

        try { System.out.println(s.find_swap(3)); } // 3
        catch (NotFound x) {}
        s.print(); // 1 5 3 0
        try { System.out.println(s.find_swap(3)); } // 2
        catch (NotFound x) {}
        s.print(); // 1 3 5 0
        try { System.out.println(s.find_swap(1)); } // 0
        catch (NotFound x) {}
        s.print(); // 1 3 5 0
        try { System.out.println(s.find_swap(7)); }
        catch (NotFound x) { System.out.println("Not found."); }

        try { System.out.println(s.find_swap(3)); }
        catch (NotFound x) {}
        s.print(); // 3 1 5 0
    }
}
```